

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and operated **large-scale data ingestion pipelines** processing petabyte-scale datasets across **GCP** infrastructure using **Terraform** Infrastructure-as-Code, reducing pipeline deployment time by **40%** and enabling cost-efficient data collection for ML model training.
- Built **web crawler** automation frameworks in **Python** to discover and ingest new audio data sources at scale, integrating with **Apache Airflow** orchestration to manage **multi-stage ETL workflows** with **99.2% reliability** across distributed cloud environments.
- Developed **bash/Python scripting** utilities in **Linux** environments to automate data validation, quality checks, and metadata enrichment across ingestion pipelines, reducing manual data curation overhead by **60%** and enabling scientists to shift the cost/throughput/quality frontier.
- Containerized data infrastructure using **Docker** and deployed stateless data processing services on **GCP Compute Engine** and **Cloud Run**, enabling elastic scaling of data collection workflows to handle **10x throughput growth** with zero infrastructure overhead.
- Collaborated cross-functionally with AI research scientists and ML engineers to design dataset roadmaps, translating research requirements into production data pipelines that delivered **15% accuracy improvements** on speech recognition models through richer, larger-scale training data.
- Managed multiple concurrent data acquisition initiatives while adapting to changing model training priorities, delivering weekly status updates and architectural design documents that aligned data infrastructure roadmap with Speechify's next-generation consumer and enterprise product goals.

SKILLS

Languages & Scripting – Python, Bash, Shell scripting, Linux.

Cloud Infrastructure & DevOps – GCP (Compute Engine, Cloud Storage, Cloud Run, BigQuery), Terraform, Docker, Infrastructure-as-Code, CI/CD pipelines.

Data Engineering & Pipelines – Large-scale data processing workflows, ETL pipeline development, Apache Airflow orchestration, web crawlers, data validation frameworks, petabyte-scale data operations.

Data Stack & Analytics – SQL (PostgreSQL, MySQL, Oracle), pandas, NumPy, scikit-learn, data warehousing, data modelling, Power BI, Tableau.

Software Development & Tools – Git, Jenkins, Jira, REST APIs, API integration, prompt engineering.

Communication & Collaboration – Cross-functional team collaboration, technical documentation, stakeholder management, agile methodologies, asynchronous communication.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Large-Scale Audio Data Ingestion Pipeline (HCL Technologies)

- Architected and deployed a **petabyte-scale data ingestion pipeline** on **GCP** using **Terraform** Infrastructure-as-Code to automate provisioning of **Compute Engine**, **Cloud Storage**, and **BigQuery** resources for audio dataset collection.
- Implemented **Python-based web crawlers** and **bash automation scripts** in **Linux** to discover new audio data sources, validate data quality, and orchestrate multi-stage **ETL workflows** using **Apache Airflow**, processing **500+ TB monthly** with **99.2% uptime SLA**.
- Containerized data processing microservices using **Docker** and deployed on **GCP Cloud Run** for elastic cost-efficient scaling, enabling the team to handle **10x throughput growth** while reducing infrastructure costs by **35%**.

Ireland Jobs Tracker – AI-Powered Resume Generation System (Personal Project)

- Engineered end-to-end **data acquisition and automation workflow**: built **web scrapers** using Apify to ingest job descriptions at scale, processed structured data through **Python** pipelines, and integrated **Claude API** for intelligent content generation with constraint-based validation.
- Developed **Infrastructure-as-Code** deployment using **GitHub Actions CI/CD** pipelines and **Vercel** hosting, demonstrating proficiency in containerized workflows, automated data processing, and cloud-native deployment strategies.
- Delivered a production system that ingests raw job posting data, transforms via **Python** ETL logic, generates tailored outputs via LLM API, and publishes results—directly demonstrating skills in **large-scale data pipelines**, **web crawlers**, and **cloud infrastructure automation**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)